

1.1

$(i, j) \in \Omega$, where $i, j \in \{1, 2, 3, 4, 5, 6\}$.

Let $A = \{\text{Second Roll is Higher than the first}\}$

First Roll	P(A)
1	5/6
2	4/6
3	3/6
4	2/6
5	1/6
6	0/6

Thus $|A| = 5 + 4 + 3 + 2 + 1 = 15$

$|\Omega| = 6 * 6 = 36$

$P(A) = \frac{|A|}{|\Omega|} = \frac{15}{36} = \frac{5}{12}$

1.5

a

$\Omega = \{1, 2, 3, 4, \dots, 40\}^5$ using the cartesian power

$\forall \omega \in \Omega P(\{\omega\}) = 1/40^5$

b

Let $3E = \{3 \text{ of } 5 \text{ numbers are even } (1-40)\}$

Let $E = 2, 4, 6, \dots, 40$

Let $O = 1, 3, 5, \dots, 39$

Since $|E| = |O|$ and $E^c = O$ then $P(E) = P(O) = 1/2$

This means we can then translate a set of 5 numbers to E, O^5 resulting in $2^5 = 32$ different E, O sets.

Therefore there are $\binom{5}{3}$ combinations of 3 even numbers. $\binom{5}{3} = \frac{5!}{3!(5-3)!} = \frac{5*4}{2} = 10$

Thus $P(3E) = 10/32 = 5/16$

1.7

a

$$P(GYG) = \frac{3}{7} \frac{4}{6} \frac{2}{5} = \frac{4}{35}$$

b

$3P(GYG) = 12/35$ this is because the probability of any order of 2 green and one yellow is the same.

1.10

a

$$\Omega = \{\infty, 1, 2, 3, \dots\}$$

$$P(\omega) = \left(\frac{5}{6}\right)^{\omega-1} * \frac{1}{6}$$

b

$$P(\omega = \infty) = \lim_{\omega \rightarrow \infty} \left(\frac{5}{6}\right)^{\omega-1} * \frac{1}{6} = 0$$

1.12

a

$$P(\omega \leq 3) = P(1) + P(2) + P(3) = \frac{1}{6} + \frac{5}{36} + \frac{25}{216} = \frac{91}{216}$$

It is also important to note that $P(1)$, $P(2)$, $P(3)$ are disjoint

b

$$P(E\omega) = 5/36 + 125/(216 * 6) + \dots$$

$$\text{Using geometric series sum: } P = \frac{5/36}{1-(5/6)^2} = \frac{5}{11}$$

1.20

a

$$\Omega = \{1, 2, 3, 4, 5, 6\}^4$$

$$P(\{\omega\}) = \frac{1}{|\Omega|} = \frac{1}{6^4}$$

b

c

1.22

1.34